

● EASY

● INFORMATION AND IDEAS

Command of Evidence

10 questions · ~15 min

05

Q1

INFORMATION AND IDEAS

Happy House is a 1920 novel by Jane Abbott. The narrator presents a young woman as being unimpressed with the house she is visiting: _____blank

Which quotation from *Happy House* most effectively illustrates the claim?

- A. “Her first feeling was of disappointment; in the square lines of the house there was little claim to beauty.”
- B. “Someone had opened one of the blinds so here there was more light.”
- C. “The door, built squarely in the middle of the house, opened almost directly upon a stone-flagged path that led in a straight line to the road.”
- D. “She tip-toed through the hall and opened the door on the right.”

Many trees produce growth rings as they age, with each ring in a tree's trunk representing one year in the tree's life. This often makes it fairly easy to determine how old a tree was when it was cut down. To do so, you look at the tree stump and count the dark rings you see. But a researcher claims that this method often can't be used to identify the age of olive trees.

Which detail, if true, would most directly support the researcher's claim?

- A. The oldest olive tree in the world is likely over 1,100 years old.
- B. Narrow growth rings can suggest that an olive tree experienced harsh conditions.
- C. Many olive trees have growth rings that are difficult to see.
- D. Olive trees thrive in areas with hot, dry summers.

Sample of Food Items from Gemini Mission
Menus

Food item	Day	Meal
Sugar cookie cubes	1	B
Chicken and vegetables	2	B
Shrimp cocktail	4	C
Hot cocoa	3	A

To make sure they got the nutrition they needed while in space, the astronauts of NASA's Gemini missions were given menus for three meals a day (meals A, B, and C) on a four-day rotating schedule. Looking at the sample of food items from these menus, a student notes that on day 1, the menu included _____blank

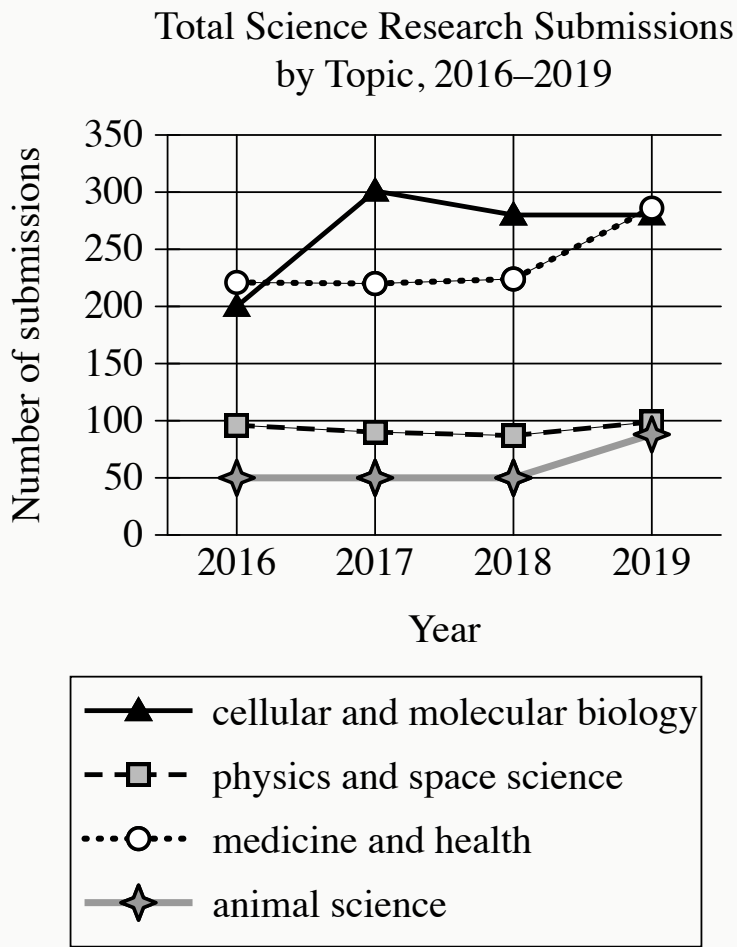
Which choice most effectively uses data from the table to complete the statement?

- A. shrimp cocktail for meal B.
- B. hot cocoa for meal C.
- C. sugar cookie cubes for meal B.
- D. chicken and vegetables for meal A.

The Milky Way galaxy is composed of millions of stars in a relatively flat structure containing a thin disk and a thick disk. Based on computer simulations and analysis of data on the brightness, position, and chemical composition of about 250,000 stars in the thick disk (collected from two telescopes, one in China and one orbiting in space), astrophysicists Maosheng Xiang and Hans-Walter Rix claim that the thick disk of the Milky Way formed in two distinct phases rather than a single one.

Which finding, if true, would most directly support the researchers' claim?

- A. The telescopes used by the researchers have detected stars of similar ages in galaxies other than the Milky Way.
- B. There's an age difference of about 2 billion years between certain stars in the thick disk.
- C. The thin disk contains about twice as many stars that can be seen from Earth as the thick disk does.
- D. The stars in the Milky Way tend to have very similar chemical compositions.



- The following 4 lines are shown:
 - cellular and molecular biology
 - physics and space science
 - medicine and health
 - animal science
- The cellular and molecular biology line:
 - Begins at 2016, 200 submissions
 - Rises sharply to 2017, 301 submissions
 - Falls gradually to 2018, 280 submissions
 - Remains level to 2019, 280 submissions
- The physics and space science line:

- Begins at 2016, 96 submissions
- Falls gradually to 2017, 90 submissions
- Falls gradually to 2018, 87 submissions
- Rises gradually to 2019, 99 submissions
- The medicine and health line:
 - Begins at 2016, 221 submissions
 - Falls gradually to 2017, 220 submissions
 - Rises gradually to 2018, 224 submissions
 - Rises sharply to 2019, 286 submissions
- The animal science line:
 - Begins at 2016, 50 submissions
 - Remains level to 2017, 50 submissions
 - Remains level to 2018, 50 submissions
 - Rises gradually to 2019, 88 submissions

A student is researching the trends in the topics submitted to a national science fair for high school students. The graph shows the number of submissions by topic that were made each year. Based on the data in the graph, the student claims that there were more medicine and health research topics submitted in 2019 than in any other year.

Which choice most effectively uses data from the graph to support the underlined claim?

- A. In 2016, the number of cellular and molecular biology topic submissions was the same as the number of animal science topic submissions.
- B. In 2019, there were more physics and space science topic submissions than there were medicine and health topic submissions.
- C. The lowest number of animal science topic submissions in a year was approximately 95 in 2016.
- D. The highest number of medicine and health topic submissions during the period shown is approximately 285 in 2019.

Average Prices Received by US Growers for Citrus Fruits, 2020–2021
(dollars per box)

Fruit	June 2020	June 2021	July 2020	July 2021
Grapefruits	\$13.80	\$23.72	\$16.13	\$22.98
Oranges	\$16.15	\$11.09	\$15.53	\$13.79
Lemons	\$19.50	\$28.94	\$21.01	\$31.78

An employee of a citrus grower in the United States is analyzing the prices of several varieties of fruit over the course of a growing season. The employee wishes to know how much a grower could have expected to receive on average for a box of lemons in July 2021. Consulting the table, the employee finds that this amount is _____blank

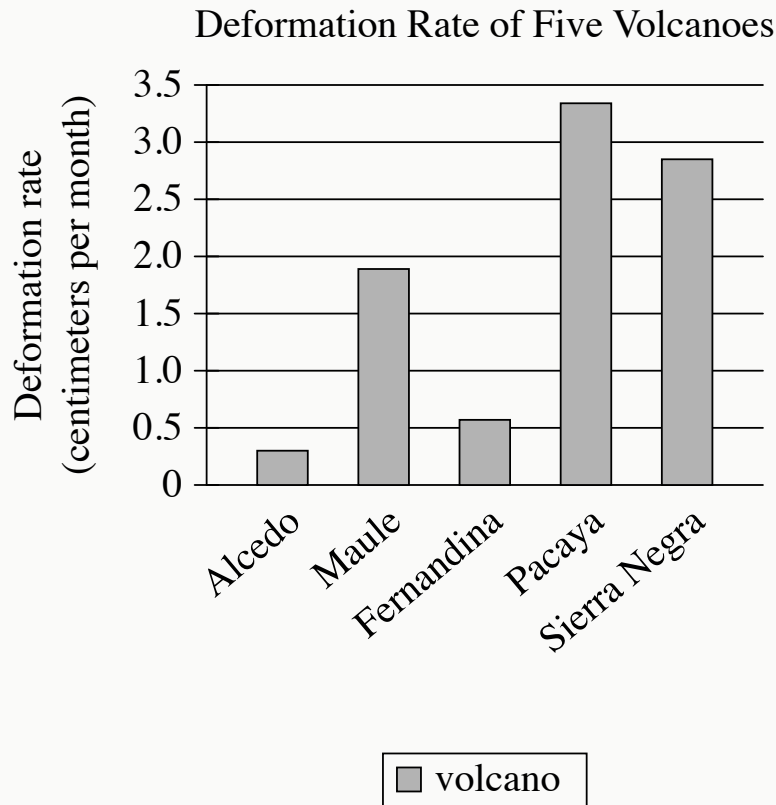
Which choice most effectively uses data from the table to complete the statement?

- A. \$31.78.
- B. \$23.72.
- C. \$22.98.
- D. \$28.94.

Roasted green chiles are a popular ingredient in Southwestern cuisine, but the traditional roasting method of burning propane is not environmentally friendly. To see if solar power could provide a better alternative, engineer Kenneth Armijo and his team roasted batches of green chiles using between 38 and 42 heliostats, which are devices that concentrate sunlight. The team was successful in reaching the same roasting temperature used in traditional propane roasting, but they found that propane yielded faster results. While the fastest solar-roasted green chiles took six minutes, batches using propane took only four. Armijo hypothesizes that they can reduce the roasting time for solar-roasted green chiles by using more heliostats.

Which finding, if true, would most directly support Armijo's hypothesis?

- A. The temperature inside the roasting drum is distributed more evenly when roasting green chiles with solar power than with propane.
- B. Attempts to roast green chiles using 50 heliostats yields results in fewer than six minutes.
- C. Green chile connoisseurs prefer the flavor of solar-roasted green chiles over the flavor of propane-roasted green chiles.
- D. The skins of solar-roasted green chiles are easier to peel than the skins of propane-roasted green chiles.

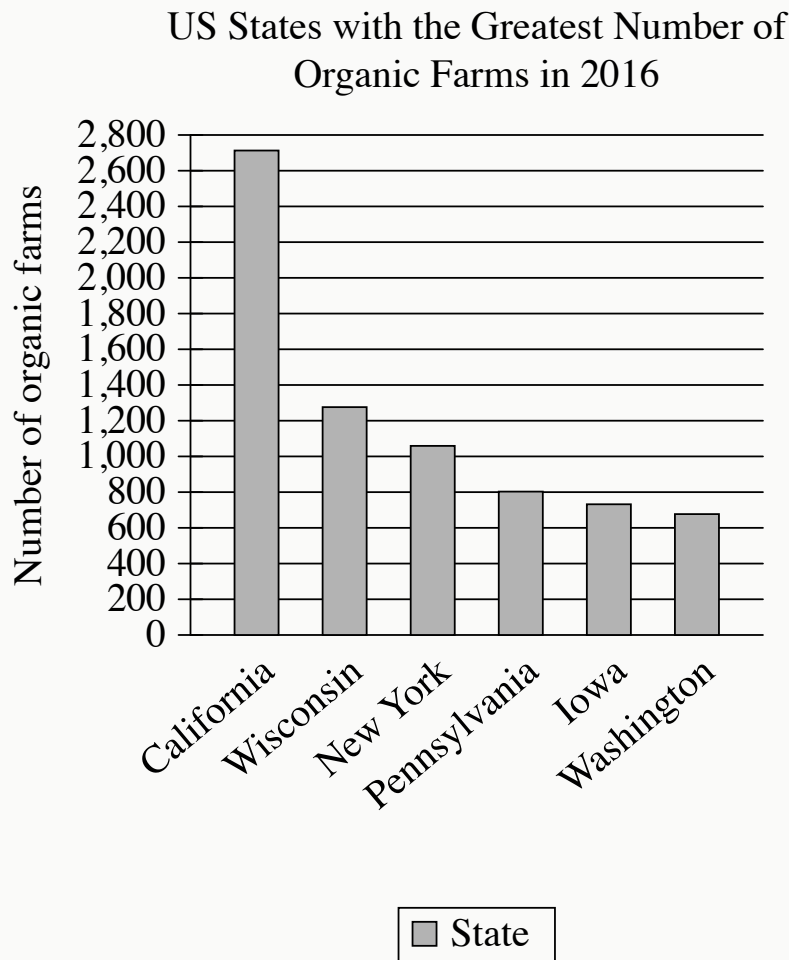


When magma moves underneath a volcano, it causes the surface of the volcano to change. This is known as deformation. Researchers recently calculated the amount of deformation occurring each month for five volcanoes in Latin America. Although Sierra Negra experienced a lot of deformation, its deformation rate was still lower than that of _____blank

Which choice most effectively uses data from the graph to complete the statement?

- A. Alcedo.
- B. Pacaya.
- C. Fernandina.

D. Maule.



- All values are approximate.
- The data for the 6 categories are as follows:
 - California: About halfway above 2,600
 - Wisconsin: Less than halfway above 1,200
 - New York: Less than halfway above 1,000
 - Pennsylvania: 800
 - Iowa: More than halfway above 600
 - Washington: About halfway above 600

Organic farming is a method of growing food that tries to reduce environmental harm by using natural forms of pest control and avoiding fertilizers made with

synthetic materials. Organic farms are still a small fraction of the total farms in the United States, but they have been becoming more popular. According to the US Department of Agriculture, in 2016 California had between 2,600 and 2,800 organic farms and _____blank

Which choice most effectively uses data from the graph to complete the text?

- A. Washington had between 600 and 800 organic farms.
- B. New York had fewer than 800 organic farms.
- C. Wisconsin and Iowa each had between 1,200 and 1,400 organic farms.
- D. Pennsylvania had more than 1,200 organic farms.

Millions of Metric Tons of Copper
Mined in 1995 and 2020

Country	1995	2020
Canada	0.73	0.59
Indonesia	0.44	0.51
Kazakhstan	0.26	0.55
Chile	2.49	5.73

While doing research for a paper about copper mining, a student finds a table with information about four different countries. The student notes that the country that mined 0.55 million metric tons of copper in 2020 had mined _____blank

Which choice most effectively uses data from the table to complete the statement?

- A. 0.59 million metric tons of copper in 1995.
- B. 0.44 million metric tons of copper in 1995.
- C. 0.26 million metric tons of copper in 1995.
- D. 2.49 million metric tons of copper in 1995.